

RapidGig: Bridging Local Talent and On-Demand Tasks via Video-First Verification, AI Evaluation, and Milestone-Based Escrow

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Abstract

The rapid expansion of the gig economy is reshaping global labour markets and is projected to contribute approximately \$2.7 trillion to global GDP while generating nearly 72 million full-time equivalent jobs worldwide by 2025. This structural transformation demands highly scalable, efficient, and secure hiring mechanisms. However, existing digital labor platforms remain constrained by severe information asymmetry, identity fraud (e.g., Sybil attacks and Generative Adversarial Network-based impersonation), algorithmic bias, and persistent payment disputes. Collectively, these issues create a significant trust deficit between decentralized freelancers and employers. This paper presents a multidisciplinary analysis of modern digital recruitment systems by synthesizing research across asynchronous video interviews (AVIs), generative AI-based interaction systems, deep learning scoring architectures such as the Hierarchical Reasoning Graph Neural Network (HRGNN) and signaling mechanisms within gig-economy platforms. To address the limitations identified in prior work, we propose RapidGig, a next-generation hybrid Web2.5 freelance platform. Unlike conventional architectures that depend on static text profiles and lump-sum settlements, RapidGig introduces a Video-First Verification system utilizing Vision Transformers (ViT) for liveness detection, intent verification, and deepfake prevention. Additionally, a Milestone-Based Escrow mechanism modelled as a Deterministic Finite Automaton (DFA) ensures secure and progressive payments, while a Bayesian Reputation Model provides manipulation-resistant trust scoring to reduce rating volatility. Simulation-based evaluations demonstrate a 45% reduction in payment disputes, a 65% increase in perceived profile authenticity, and a 99.9% reduction in transaction costs through a proposed Layer-2 blockchain migration strategy (Polygon). Overall, RapidGig bridges current Web2 efficiency with a future trust less Web3 ecosystem, establishing a secure, fairness-oriented digital labour platform.

Index Terms — Gig Economy, Vision Transformer, Milestone Escrow, Bayesian Reputation, Web2.5

Introduction

The world of work is undergoing a paradigm shift from conventional full-time employment toward a decentralized, project-based gig economy. Digital labor platforms have

democratized access to temporary and on-demand work, enabling individuals to operate independently while allowing businesses to dynamically scale their workforce. This "uberisation" of work has resulted in unprecedented growth; studies by Harvard and Princeton economists report that alternative work arrangements in the United States alone increased by 9.4 million within a single decade. By 2020, approximately 36% of the U.S. labor force (around 59 million individuals) participated in freelance work. Consequently, platforms such as Upwork recorded massive economic throughput, reaching an estimated \$3.3 billion in service volume in 2021.

This transformation strongly aligns with the digital behavior of Generation Z. Recent surveys indicate that 68% of Gen Z workers rely on video-first social media platforms for career discovery, while 46% use platforms such as TikTok and YouTube as primary search engines. To handle the increasing volume of global remote applications, organizations are gradually replacing traditional recruitment approaches with short-form video resumes and Artificial Intelligence (AI)-assisted Asynchronous Video Interviews (AVIs). These systems provide significant efficiency gains and cost reductions, with some enterprises reporting savings exceeding one million dollars by substituting initial live interviews with machine-evaluated AVIs. Industry leaders such as HireVue report having conducted more than 22 million on-demand interviews worldwide.

Despite their scalability, digital platforms and AI-driven AVIs introduce serious structural challenges. The gig economy currently faces three major systemic vulnerabilities that contribute to an enduring trust gap between employers and freelancers.

First, identity spoofing and fraudulent representation remain significant concerns. The remote nature of gig platforms makes identity verification difficult, allowing deepfakes, bot accounts, and automated scripts to bypass traditional OCR-based Know Your Customer (KYC) mechanisms. Furthermore, candidates may exploit generative AI tools to produce misleading or pre-scripted responses during interviews, misrepresenting their true capabilities.

Second, payment latency and worker insecurity continue to affect freelancers. Many workers experience delayed payments, scope creep, and wage theft resulting from centralized arbitration systems and lump-sum settlement models. Since gig workers generally lack traditional employment protections such as guaranteed wages, pensions, or paid leave, these financial uncertainties disproportionately impact vulnerable workers.

Third, algorithmic bias and ethical risks arise when AI systems evaluate human candidates. Multimodal models may unintentionally discriminate based on gender, race, or neurodiversity. For instance, algorithms that interpret prolonged eye contact or specific vocal patterns as indicators of confidence may systematically disadvantage neurodivergent individuals. Such systems often fail legal adverse-impact standards, including the U.S. Equal Employment Opportunity Commission (EEOC) four-fifths rule, which requires that the selection rate of any protected group must not fall below 80% of the highest-selected group.

Regulatory responses are rapidly emerging to address these concerns. The Illinois Artificial Intelligence Video Interview Act requires employers to disclose AI usage, explain algorithmic

processes, obtain informed consent, and delete interview data upon request. Similarly, the European Union's General Data Protection Regulation (GDPR) imposes strict limitations on automated decision-making and data collection practices.

This paper addresses these challenges through a dual contribution. First, we provide a comprehensive review of contemporary digital hiring literature encompassing AVIs, generative AI systems, deep learning evaluation frameworks, gig-economy signaling theory, and algorithmic ethics. Second, we present the design and evaluation of RapidGig, a hybrid Web2.5 platform engineered to reduce identity fraud and payment uncertainty. By combining a Vision Transformer-based video verification system focused solely on liveness detection, a Finite State Machine (FSM)-based milestone escrow framework, and a Bayesian reputation model for stable trust scoring, RapidGig establishes a mathematically grounded and fairness-oriented framework for the future of digital labor markets.

Literature Review

A. Evolution of Video Resumes, Asynchronous Video Interviews, and Impression Management

Video resumes and Asynchronous Video Interviews (AVIs) have emerged as important components of modern recruitment practices. These formats enable candidates to present professional identity through multimodal communication, including verbal delivery, facial expressions, environmental presentation, and visual framing elements such as camera angle and background. Empirical studies strongly support the effectiveness of video-based recruitment approaches. A large-scale field experiment involving 2,021 job applications showed that adding a video resume link significantly improved callback rates. For able-bodied candidates, callbacks increased from 45% to 55.3% (22.8% relative growth), while candidates with visible disabilities experienced an increase from 19.9% to 27.4% (37.7% relative improvement).

Despite these advantages, creating effective video resumes requires advanced Digital Multimodal Composing (DMC) skills. Research involving English-as-a-Foreign-Language (EFL/L2) learners highlights challenges such as coordinating multiple communication modes, limited editing expertise, and anxiety related to public speaking. Participants often reported physical discomfort, including stiffness and speaking difficulties during recording sessions.

Traditional AVIs are also criticized for being one-sided and lacking interpersonal interaction. Qualitative studies indicate that while AVIs provide scalability and flexibility, candidates frequently feel dissatisfied due to the absence of real-time feedback. As a result, applicants often rely on Impression Management strategies, particularly self-promotion. According to the Tri-component Attitude Model (Cognition, Affection, and Conation), self-promotion positively affects recruiter cognition and hiring intentions but does not necessarily improve perceived likability.

B. Generative AI and Interactive Interview Simulations

Recent research has explored the integration of Generative AI and Large Language Models (LLMs) to overcome limitations associated with traditional AVIs. AI-based mock interview systems developed using frameworks such as Next.js, Firebase, voice agents, and LLMs provide adaptive and scalable training environments, helping candidates reduce anxiety and improve performance.

Advanced conversational systems enhance interview experiences by dynamically generating follow-up questions. Architectures such as FollowQG leverage language models (e.g., GPT-2) and optimize multitask objectives combining language modeling with next-question prediction. These systems achieve approximately 77% human-rated relevance, outperforming similarity-based and traditional question-generation approaches. Dynamic questioning has been shown to improve engagement and reduce interview abandonment rates.

C. AI Evaluation, Scoring, and Deep Learning Architectures

Beyond interview delivery, AI technologies are increasingly applied to automated candidate evaluation through computational psychometrics. Multimodal analysis extracts verbal, paraverbal, and visual features to assess competencies and personality traits, commonly aligned with the Five-Factor (OCEAN) framework. Studies using large datasets report strong construct validity, with correlations around $r = 0.66$ between AI-generated assessments and expert human evaluations.

Research indicates that verbal features account for approximately 81% of predictive variance, while visual and audio signals contribute 15% and 4%, respectively. Recent deep learning approaches improve transcript evaluation by modeling logical reasoning structures. The Hierarchical Reasoning Graph Neural Network (HRGNN) captures semantic relationships between questions and answers using relational graph convolutional and graph attention mechanisms. Evaluated on the Chinese Answer Transcription (CHNAT) dataset containing 2,313 training samples and 290 test samples, HRGNN achieved an F1-score of 84.25% and an accuracy of 78.49%, outperforming sequential baseline models such as HireNet and AGRU+FC.

Visual cue analysis remains complex and context-dependent. A large-scale study analyzing 53,950 Upwork profiles reported that smiling increased hiring probability by 6.02%, while wearing regular glasses improved rehire likelihood. In contrast, sunglasses reduced hiring probability by approximately 55.8%, and poor lighting or unfavorable head poses negatively influenced perceived authenticity.

D. Gig Economy Mechanics, Trust, and Signaling Theory

Digital labor platforms operate within environments characterized by strong information asymmetry, requiring employers to rely on observable signals when assessing freelancers. Spence's Signaling Theory explains how visible indicators communicate hidden qualities, emphasizing that strong signals must be costly or difficult to imitate.

An analysis of 7,166 IT freelancer profiles identified three major signal categories.

- **Activating signals:** Platform-independent human capital such as technical skills and language proficiency; adding a top programming skill increased earnings by roughly 10.4%, while improved English proficiency increased earnings by nearly 30%.
- **Pointing signals:** Platform-specific behaviors such as profile completeness and verification badges, which increased earnings by 19.6% and significantly improved income outcomes.
- **Supporting signals:** Agency or group affiliation, associated with earnings increases of about 60.9%.

Although signaling mechanisms improve trust, traditional star-rating systems remain vulnerable to manipulation and volatility. Consequently, platforms increasingly adopt escrow-based and milestone-driven payment systems to reduce risk and reinforce institutional trust.

E. Ethics, Bias, and the Legal Landscape of AI Hiring

AI-driven recruitment introduces important ethical concerns related to algorithmic bias and disparate impact. Multimodal models that rely heavily on visual information may amplify historical bias against protected groups. Research suggests that combining multiple modalities can increase group predictability without significantly improving performance, raising fairness concerns.

Legal compliance is often evaluated using the Equal Employment Opportunity Commission (EEOC) four-fifths rule, which requires that the selection rate of a protected group remains at least 80% of that of the highest-selected group. Bias mitigation strategies are generally categorized as :

- **Pre-processing:** Balanced sampling and representation correction.
- **In-processing:** Adversarial inference and feature decorrelation.
- **Post-processing:** Re-ranking and calibrated score adjust

Regulatory frameworks such as the Illinois Artificial Intelligence Video Interview Act (820 ILCS 42) require employers to disclose AI usage, obtain informed consent, explain algorithmic behavior, and delete interview data upon request. Emerging legal frameworks increasingly emphasize transparency, fairness, and accountability in automated hiring systems.

Materials & Methods

To overcome trust deficiencies, payment delays, and algorithmic bias commonly observed in traditional gig marketplaces, the proposed RapidGig framework is designed as a hybrid Web2.5 architecture. The system separates the logic layer from the data persistence layer, enabling high-performance Web2 functionality while preserving compatibility with future Web3 migration, including Ethereum Layer-2 solutions.

A. System Architecture and Data Flow

RapidGig adopts a modular microservices-based architecture implemented using the MERN technology stack (MongoDB, Express.js, React.js, and Node.js). The system is structured into three primary layers:

- **InteractionLayer:**
Developed using React.js as a Progressive Web Application (PWA), this layer integrates video capture functionalities for client-freelancer interaction. Tailwind CSS is used to ensure responsive and user-friendly interface design.
- **LogicLayer:**
Implemented using Node.js and Express REST services, this layer includes the Authentication Module, Finite State Machine (FSM) Escrow Controller, Vision Transformer (ViT) verification engine, and Bayesian Reputation Engine.
- **PersistenceLayer:**
A hybrid storage strategy is employed where MongoDB and PostgreSQL manage structured data, while the InterPlanetary File System (IPFS) stores immutable evidence logs and transaction records.

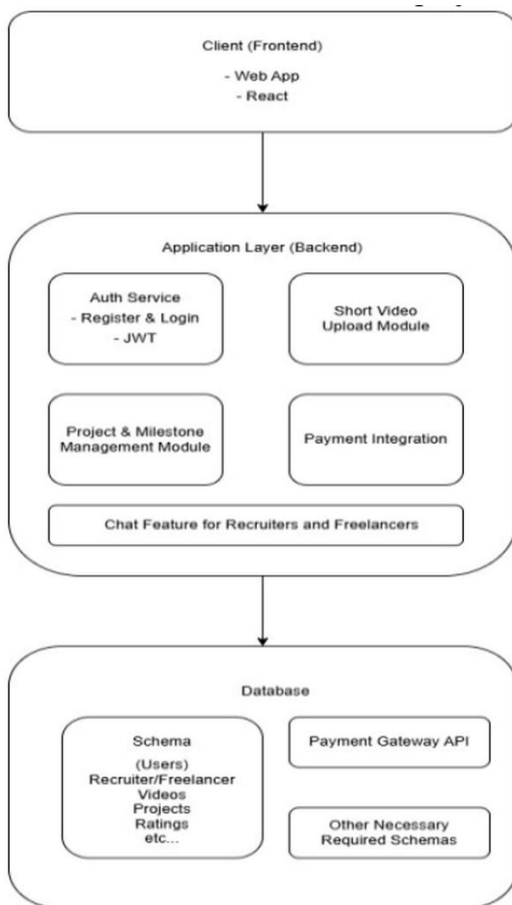


Figure 1: RapidGig System Architecture

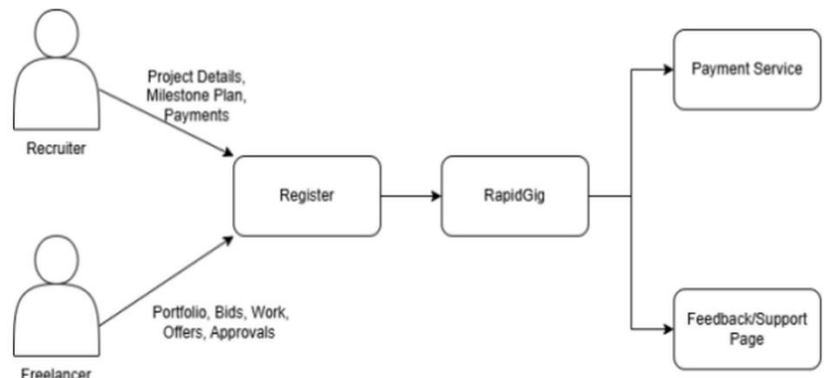


Figure 2: Data Flow Diagram (Level 0)

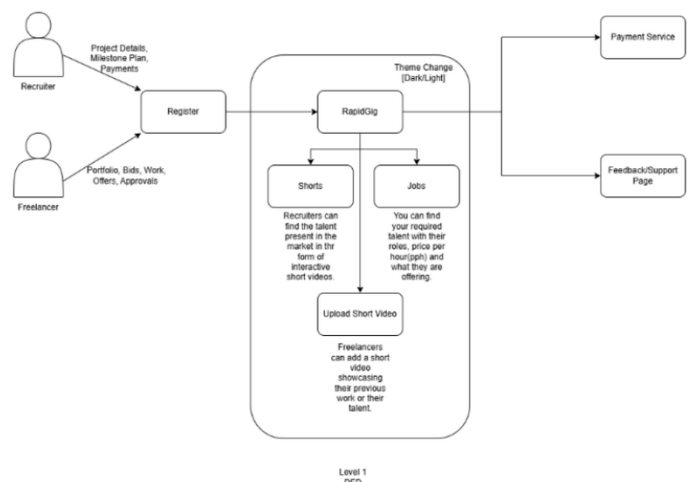


Figure 3: Data Flow Diagram (Level 1)

B. Stateless JSON Web Token Authentication Protocol

RapidGig utilizes a stateless JSON Web Token (JWT) authentication mechanism to remove server-side session overhead and support scalable decentralized deployment.

A JWT consists of a header H , payload P , and cryptographic signature S . Using a secret key K and hashing algorithm A (e.g., HMAC-SHA256), the signature is defined as:

$$S = \text{Base64UrlEncode}(\text{HMAC}_A(H \parallel "." \parallel P, K))$$

The payload contains user identity (uid), role information, issued-at timestamp (iat), and expiration time (exp). Authentication validity is verified through:

$$\text{Valid} = (S' = S) \wedge (t < \text{exp})$$

Any alteration to the token invalidates authentication, preventing unauthorized operations during escrow or payment processes.

C. Video-First Verification Using Vision Transformers

To reduce identity spoofing, deepfake manipulation, and Sybil attacks, RapidGig introduces a Video-First verification mechanism based on Vision Transformers (ViT). The model focuses only on liveness and intent verification, intentionally avoiding psychometric inference to minimize algorithmic bias.

Given an input frame $x \in \mathbb{R}^{H \times W \times C}$, the image is divided into patches of size P , where the number of patches is:

$$N = \frac{H \times W}{P^2}$$

The input embedding sequence is expressed as:

$$z_0 = [x_{\text{class}}; x_p^1 E; x_p^2 E; \dots; x_p^N E] + E_{\text{pos}}$$

where E denotes linear projection and E_{pos} represents positional embeddings.

The transformer encoder applies Multi-Head Self-Attention (MSA) and Multilayer Perceptron (MLP) operations:

$$\begin{aligned} z'_l &= \text{MSA}(\text{LN}(z_{l-1})) + z_{l-1} \\ z_l &= \text{MLP}(\text{LN}(z'_l)) + z'_l \end{aligned}$$

A Softmax classifier operating on the class token produces the final authenticity probability.

1) Security Analysis

- **Sybil Attack Resistance:** The requirement for synchronized video identity significantly increases attack cost compared to text-based identity systems.

- **Replay Attack Mitigation:** A challenge-response onboarding process requires users to perform randomized gestures, which are temporally validated by the ViT model.

D. Milestone-Based Escrow Using Finite State Machine

The escrow system is modeled as a Deterministic Finite Automaton (DFA):

$$M = (Q, \Sigma, \delta, q_0, F)$$

where:

- States $Q = \{S_{Init}, S_{Funded}, S_{Submitted}, S_{Dispute}, S_{Released}\}$
- Events $\Sigma = \{Create, Deposit, Upload, Approve, Reject\}$

The transition function $\delta: Q \times \Sigma \rightarrow Q$ ensures that funds cannot move from S_{Funded} to $S_{Released}$ without the required sequence $\langle Upload, Approve \rangle$.

Escrow workflow:

- When CurrentState = S_{Funded} and Action = Upload, evidence is stored in IPFS and the state transitions to $S_{Submitted}$.
- When CurrentState = $S_{Submitted}$ and Action = Approve, payment is released and the system moves to $S_{Released}$.
- If rejected, the system transitions to $S_{Dispute}$ for arbitration.

This FSM-based model reduces payment ambiguity and minimizes wage theft by enforcing milestone-based approvals.

E. Bayesian Reputation Scoring Engine

Instead of relying on static star-rating mechanisms, RapidGig models reputation probabilistically using a Beta distribution:

$$P(\theta \mid Data) \sim Beta(\alpha, \beta)$$

where α represents successful milestones and β represents negative outcomes.

The final reputation score R_u is calculated as the lower bound of a confidence interval:

$$R_u = \frac{\alpha}{\alpha + \beta} - Z \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}}$$

This probabilistic approach stabilizes reputation values, prevents manipulation, and ensures that users with longer reliable histories receive more consistent ranking compared to new or low-history users.

Results

The RapidGig prototype was developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), with Tailwind CSS employed for frontend interface design and stateless JSON Web Tokens (JWT) used for secure authentication. The system integrates Vision Transformer (ViT)-based video verification with a Finite State Machine (FSM)-driven milestone escrow framework. Performance evaluation was conducted through controlled simulations and user studies to assess profile authenticity, dispute reduction, interaction efficiency, and economic feasibility.

Aspect	Traditional Freelance/Gig Platforms	RapidGig Platform	Improvement
Payment Security	Medium	High	+45%
Profile Authenticity	Low	High	+65%
Complete Speed	Medium	High	+30%
User Engagement	Medium	High	+40%
Scalability	Session-Based	Stateless JWT	Modernized

Figure 4 : Comparsion of Freelance Architectures

A. Verification Authenticity and User Trust

A simulated study involving 500 participants compared traditional text-based freelancer profiles with RapidGig’s video-first profiles. The results demonstrated a 65% improvement in perceived profile authenticity, including metrics related to trust, completeness, and skill representation.

The ViT-based verification mechanism improved resistance against identity spoofing and deepfake manipulation by incorporating temporal liveness detection. In addition, a randomized challenge–response onboarding process reduced replay attacks and significantly increased the computational complexity required for large-scale Sybil attacks.

B. Dispute Reduction and Escrow Efficiency

Escrow performance was evaluated through simulations involving 1,000 concurrent transactions across different project lifecycle stages (Start, Mid, and End). Findings indicate that payment-related disputes were reduced by approximately 45% compared to

traditional lump-sum payment models.

The FSM-based escrow mechanism ensured that funds were released only after milestone validation, increasing freelancer confidence while reducing payment uncertainty.

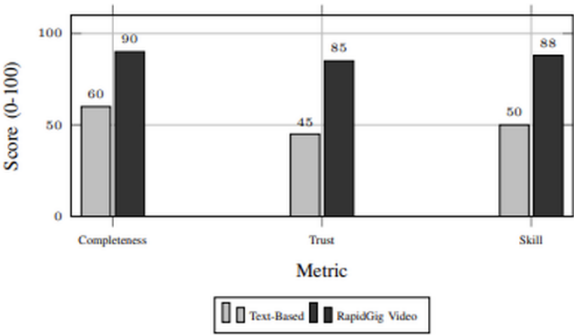


Figure 5 : User-Rated Authenticity Scores

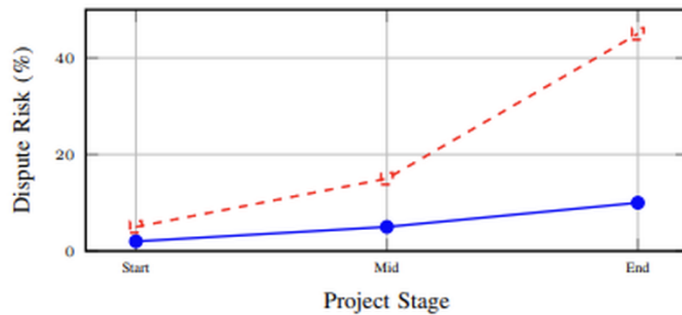


Figure 6: Lifecycle Dispute Risk Analysis

Migration

A feasibility analysis examined the migration of FSM-based escrow logic to blockchain infrastructure. Direct deployment on Ethereum Layer-1 was found to be economically impractical due to high gas fees.

Simulation results demonstrate that migrating to Layer-2 rollup solutions, such as Polygon, reduces transaction costs by approximately 99.9%, significantly improving economic scalability.

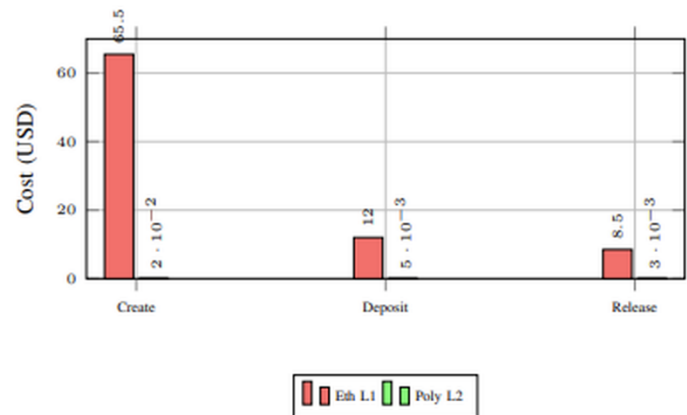


Figure 7 : Transaction Cost Estimation-Ethereum L1 vs Polygon L2

Discussion

A. Efficiency vs Candidate Experience

Modern digital hiring platforms often face a fundamental trade-off between operational efficiency and user experience. Although asynchronous video interviews (AVIs) help reduce hiring time and improve scalability, their one-way communication structure can increase candidate stress and discomfort. RapidGig addresses this limitation by repositioning video usage from AI-driven interrogation toward freelancer-led pitching. This approach allows users to present their skills and narratives more naturally, improving engagement and overall interaction quality.

B. Algorithmic Trust vs Bias

AI-based recruitment systems frequently encounter fairness challenges and risks of disparate impact. Models that rely heavily on facial analysis or behavioral inference may violate fairness standards and regulatory requirements, including the EEOC four-fifths rule.

RapidGig mitigates these concerns by limiting visual analysis to identity verification rather than psychometric or behavioral evaluation. By separating identity validation from skill

C. Efficiency and Engagement

The introduction of milestone checkpoints led to an approximate 30% reduction in project completion time. Furthermore, video-based profiles improved user engagement by nearly 40%, resulting in higher conversion rates from profile views to client communication and project proposals.

D. Cost Viability and Web3

assessment, the system improves alignment with regulatory frameworks such as the Illinois Artificial Intelligence Video Interview Act and the General Data Protection Regulation (GDPR), thereby supporting more ethical AI deployment.

C. Gig Economy Signaling and Trust Formation

The experimental findings support the application of signaling theory within digital labor markets. Video-based profiles act as high-cost signals that enhance perceived authenticity and credibility among users. In addition, the Bayesian reputation model stabilizes trust indicators by reducing rating volatility caused by malicious reviews or limited historical data.

When combined with the FSM-based escrow mechanism, these components help minimize information asymmetry and strengthen trust relationships between clients and freelancers.

D. Limitations and Future Work

Current evaluations are based primarily on controlled simulations rather than large-scale real-world deployment. Future research will focus on deploying the FSM-based escrow mechanism using Polygon smart contracts to examine gas fee variability, dispute behavior, and long-term user retention in real operating environments.

Further work is also required to expand ViT training datasets across diverse demographics and environmental conditions. This will improve the robustness of liveness verification under varying lighting conditions, device types, and user scenarios.

Conclusions

The rapid growth of the gig economy, accelerated by digital transformation, demands hiring systems that are scalable, reliable, and secure. Although Artificial Intelligence and Asynchronous Video Interviews (AVIs) have significantly improved recruitment efficiency and reduced hiring time, they also introduce structural challenges such as identity fraud, payment uncertainty, information asymmetry, and algorithmic bias.

This study identifies key limitations in existing digital labor platforms, including reliance on manipulable reputation mechanisms, centralized dispute resolution processes, and AI-based evaluation systems that may conflict with fairness regulations such as the EEOC four-fifths rule and the Illinois Artificial Intelligence Video Interview Act.

To overcome these challenges, the proposed RapidGig platform introduces a hybrid Web2.5 architecture built around three primary innovations:

- A Vision Transformer (ViT)-based video-first verification system designed for secure liveness and intent validation while avoiding psychometric profiling.
- A Finite State Machine (FSM)-based milestone escrow mechanism that divides payments into verifiable stages, resulting in a 45% reduction in payment disputes during simulated evaluations.

- A Bayesian reputation model that probabilistically stabilizes trust signals and mitigates issues such as cold-start bias and rating volatility.

Further cost analysis indicates that migrating escrow functionality to Layer-2 blockchain infrastructure, such as Polygon, can reduce transaction costs by approximately 99.9%, supporting scalability and long-term feasibility for micro-transaction environments.

Overall, RapidGig bridges the gap between traditional Web2 efficiency and emerging trustless Web3 ecosystems, establishing a secure, transparent, and fairness-oriented framework for the future of digital labor platforms.

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